



Introduction to Bootstrap

1. Need of bootstrap
2. Adding bootstrap to HTML pages
3. Basic container div
4. Bootstrap grid system

- Introduction
 - It is free, open source and popular front-end development framework
 - Makes it possible for a web page to detect the visitor's screen size and orientation and automatically adapt the display accordingly.
- Brief history
 - Bootstrap was initially released as a project within Twitter in 2010
 - Bootstrap was open-sourced accessible to web developer community in 2011
- Version
 - latest stable version of Bootstrap as of August 27, 2024 is Bootstrap 5.3.

- Advantages
 - Save lots of time : using the Bootstrap predefined design templates and classes
 - Responsive features : websites that appear more appropriately on different devices without disturbing the content.
 - Consistent design: the design and layout of the web pages will be consistent.
 - Easy to use : basic working knowledge of HTML, CSS and JavaScript is sufficient
 - Compatible with browsers : with all modern browsers such as Chrome, Firefox, Safari, Internet Explorer, etc.
 - Open Source : Bootstrap library is completely free to download and use.

- Following steps are used to make HTML page use the functionality of bootstrap library.
 1. Add the HTML5 doctype
 2. Add the <meta> tag inside the <head> element to ensure proper rendering and touch zooming
 3. Adding bootstrap files to HTML page :
2 ways to start using bootstrap 5 on the web page similar like jquery:
 - i. Include Bootstrap 5 from a CDN
 - ii. Download Bootstrap 5 library
 4. Bootstrap 5 requires a containing element to wrap site contents

- Containers are basically used to wrap content with some padding.
- It is compulsory if grid system to be used
- Fixed Width Container
 - fixed-width container that adjusts its width based on the screen size.
- Fluid Container
 - full width container, that will always span the entire width of the screen.
- Responsive Containers
 - Classes like `.container-sm | md | lg | xl` are used to specify when the container should be responsive.

- Adding Background and Borders to Containers
 - container doesn't have any background-color or border. It can be applied like
`<div class="container bg-dark text-white">`
- Applying Paddings and Margins to Containers
 - Containers have padding of 12px on the left and right sides, and no padding on the top and bottom sides. To apply extra padding and margins
`<div class="container border py-3 my-3">`

- Basics
 - It is basically used to generate columnar output.
 - It uses twelve column system
 - Few columns can be grouped together to create wider columns.
- Rows
 - To create single row of columns use `<div class="row">`
- Columns
 - six-column class available which can adapt to diff screen sizes
- Responsive columns
 - the `.col` or `.col-*` class can be used for same size columns on all devices
- Column positioning
 - Possible to have blank spaces between 2 columns by using `offset-*-*`